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CLINICAL AND EPIDEMIOLOGICAL ASPECTS OF HEMORRHAGIC FEVER WITH RENAL SYNDROME (HFRS) IN GREECE

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Twenty-three cases of hemorrhagic fever with renal syndrome were serologically diagnosed in rural populations of Greece between 1982 and 1985. The clinical picture of the disease was severe, (3 deaths recorded among 23 cases) and more closely resembled Korean hemorrhagic fever than nephropathia epidemica. Examination of patients' sera by indirect immunofluorescence (IFA) assays and plaque reduction neutralization (PRNT) tests using Hantaan 76-118, Puumala, and Seoul (Urban rat-associated Hantaan-like) viruses, suggested that the virus existing in Greece is either prototype Hantaan virus or a distinct virus which is closely related to Hantaan. Small mammal survey results suggest that *Apodemus flavicollis* may be the host of this virus. Serosurvey results of 2,653 blood samples taken from apparently healthy residents (farmers, shepherds, and woodcutters) from 22 of 54 counties in Greece, showed the percentage of seropositives to be 4%, with a range from 0 to 14%. Fifteen of 22 counties were characterized as endemic areas of the virus due to the presence of seropositive individuals. In 6 of the 15 endemic areas, overt human disease was serologically diagnosed.

INTRODUCTION

Hemorrhagic fever with renal syndrome (HFRS) is an acute, febrile nephropathy caused by several viruses of the genus *Hantavirus* of the family Bunyaviridae (17, 12). Today three distinct viruses of this genus are known to cause the disease in humans: Hantaan virus, the cause of Korean hemorrhagic fever in Korea and epidemic hemorrhagic fever in China, Puumala virus, the cause of nephropathia epidemica in Scandinavia, and Seoul virus, (Urban rat-associated Hantaan-like virus), the cause of a less severe form of HFRS in China and Korea (8). All of these viruses are associated with specific rodent hosts: prototype Hantaan, Puumala, and Seoul viruses are associated with *Apodemus agrarius*, *Clethrionomys glareolus*, and domestic rats, respectively. Another isolate Prospect Hill virus recovered in the USA from *Microtus pennsylvanicus*, is not presently associated with human disease (9).

Isolated cases of HFRS have recently been reported in several European countries outside Scandinavia, including France (3), Scotland (16), Belgium (15), Czechoslovakia (10), Hungary (14), and the USSR (2). In the Balkan Peninsula, clinical cases of the disease have been reported from Bulgaria (6), Yugoslavia (5), and Rumania (4).

In 1981, antibodies to Hantaan virus were reportedly found in sera from residents of northern

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Greece (11). In 1982, two cases of HFRS were serologically diagnosed, and foci of the virus transmission were found in different parts of northern Greece (1). In 1983, an outbreak of the disease appeared in northwestern Greece (Epirus State) resulting in eight cases, one of which was fatal (13). This paper describes the clinical manifestations and symptoms of 23 serologically confirmed HFRS cases (10 previously reported) in Greece. Clinical symptoms of these cases are compared with those of both Korean hemorrhagic fever and nephropathia epidemica. Several endemic areas of Greece (9 previously reported) are identified, and serological data concerning the rodent host of the virus is presented. Finally, serological findings are presented which indicate that the virus existing in Greece is either identical to the prototype 76-118 strain of Hantaan virus or is a distinct virus closely related to Hantaan.

MATERIALS AND METHODS

Patients

Acute and convalescent sera were collected between 1982 and 1985 from patients whose illnesses had been clinically diagnosed as leptospirosis, acute nephritis, or acute renal insufficiency, and from patients with influenza-like diseases or pyrexia of unknown origin. The patients were residents of various parts of northern Greece, primarily Thrace, Macedonia, Thessalia, and Epirus. Patients were referred to the local hospitals (County General Hospitals) or the General hospitals of Thessaloniki for specialized diagnosis and treatment. The patient's serum, single or paired, was examined on the day of arrival in the laboratory, or stored at —20°C until it could be tested by the indirect immunofluorescence (IFA) test. Sera were identified by age, sex, occupation, previous travel history (mainly abroad), residence, and medical record of the patient. Additionally, 25 paired blood samples were drawn from the staff of a hospital ward in which 5 HFRS patients had been treated, and 30 single blood samples were obtained from the wives and children of 5 HFRS patients in order to investigate possible man-to-man transmission.

Human serosurvey

Sera from 2,653 (792 previously examined) subjects were obtained in 22 of the 54 counties of Greece: 13 in northern Greece (Thrace, Macedonia, and Epirus States), 4 in central Greece (Thessalia State), 2 in southern Greece (Peloponnisos State), 1 in Corfu Island and 2 in Crete Island. Sera were collected annually from 1980 to 1985 and stored at —20°C until they could be tested for anti-Hantaan virus antibody by IFA test. These sera were identified by age, sex, occupation, previous travel history (mainly abroad), and residence of the donor.

Small mammal collection

Small mammals were live-trapped in the fields and forests surrounding the villages in which HFRS cases had been diagnosed. Rats were captured using Tomahawk live traps (Tomahawk, Wisconsin, USA), and smaller mammals using Swedish live traps (Algren AB, Hillerstorp, Sweden). All traps were baited with a mixture of peanut butter, rolled oats, and fresh apples. Captured animals were processed at temporary field stations or in the laboratory in Thessaloniki. In either case, whole blood was aspirated from mice with capillary tube Pasteur pipettes following rupture of the retro-orbital sinus. Rats were anesthetized with 0.1 ml intramuscular injection of Ketamine, then exsanguined by cardiac puncture. Blood was allowed to clot at room temperature and then centrifuged. Serum was removed and stored at —20°C until it could be tested by IFA test as described below. Lungs, spleen, and kidneys were aseptically removed and stored individually in plastic vials containing liquid nitrogen. Captured animals were identified by species. Additionally, study skins and skulls were prepared from selected *Apodemus* sp. rodents, and specific identification took place at the United States National Museum of Natural History, Smithsonian Institution, Washington, D.C..

Serological techniques

Patients' sera, and apparently healthy individual's sera collected for serosurvey, were examined by IFA tests with goat anti-human fluorescence immunoglobulin. Small mammals' sera were tested by the same method with rabbit anti-mouse fluorescence immunoglobulin (Gibco Diagnostic, Madison, Wisconsin, USA). Spot-slides contained Vero E-6 cells, approximately 50% infected with the 76-118 strain of Hantaan virus. IFA test is known to be widely cross-reactive and detects circulating antibodies to Hantaan, Puumala, and Seoul viruses. Human sera were considered positive if characteristic cytoplasmic fluorescence was detected at a 1:16 dilution, while rodent sera were considered as positive at 1:32 dilution.

Selected sera were further tested by plaque reduction neutralization tests (PRNT). PRNT was performed in parallel with prototype 76-118 strain of Hantaan virus, Puumala virus, and Seoul virus on Vero E-6 cells grown in 25 ml flasks. Equal volumes of diluted serum and 200 plaque forming units (PFU) of the corresponding virus were mixed and incubated overnight at 4°C. The mixtures were inoculated onto confluent monolayers

of Vero E-6 cells and allowed to adsorb for 1 hour at 37°C with periodic rocking, then covered with a nutrient 1% agarose overlay and incubated at 37°C for approximately 10 days. They were stained with a dilution of neutral red solution, and plaques were counted at 24-48 hours poststaining. The titer was recorded as the reciprocal of the greatest dilution reducing 50% of the plaque dose. Appropriate positive and negative serum controls were included in both IFA and PRNT tests.

Virus isolation attempts

Lung tissues from captured rodents were homogenized individually in a mechanical blender (Stomacher Blender, Model 80, Tekmar Co., Cincinnati, Ohio, USA), and virus isolation attempted following procedures previously described (9).

RESULTS

Patients and disease

One-hundred-eighty male and female farmers, shepherds, and woodcutters were admitted to various General Hospitals of Thessaloniki and other General Hospitals located in the county capitals, with clinical diagnoses of leptospirosis, acute nephritis, or acute renal insufficiency. The diagnosis of HFRS was serologically confirmed in 23 of these patients by rising antibody titers (IgM and IgG) to Hantaan virus. In cases where only single blood samples were available, the determination of specific IgM antibodies to Hantaan virus in high titers confirmed the diagnosis. None of the 110 patients with influenza-like disease or pyrexia of unknown origin were found to be infected by Hantaan virus.

Analysis of data concerning clinical signs and symptoms of the disease as reported in the patient's medical records is shown in Table 1. Of 23 serologically diagnosed cases, 3 (14%) died and 12 (54%) developed severe symptoms including flushing over face and neck, conjunctival injection, pneumonic infiltration, pulmonary edema, confusion, shock, and hemorrhagic manifestations. Eight of these severely ill patients (35%) required renal dialysis. The predominant symptoms in all patients were fever, headache, nausea, vomiting, and abdominal pain, while flushing of the face, conjunctival injection, pulmonary edema, shock, and hemorrhagic manifestations were only common in the severely ill patients. Proteinuria with microscopic haematuria and increased serum urea and creatinine were present in all patients. Additionally, one inapparent infection was serologically diagnosed in the wife of a severely ill patient by detection of high IgG (1:1024) and IgM (1:512), against Hantaan virus. Table 2 shows a comparison of the clinical characteristics of HFRS patients from Greece, Finland, and Korea.

TABLE 1.
Clinical symptoms and signs in 23 HFRS Greek patients.

Symptoms and Signs	No. of Patients
Fever	23
Rigors	23
Headache	23
Malaise	23
Abdominal pain	22
Myalgia	21
Arthralgia	19
Vomiting	19
Backache	18
Flush over the face and neck	15
Conjunctival injection	15
Hypotension	15
Dulled sensorium	10
Confusion	10
Shock	5
Diarrhea	4
Pneumonic infiltration	3
Cough	3
Hemorrhagic manifestations	3
Pulmonary edema	2

TABLE 2.
Comparison of clinical characteristics of HFRS patients in Greece, Finland, and Korea.

	Percentages of cases		
	Greece	Finland*	Korea***
Hemorrhagic manifestations			
Conjunctival injection	65	18	97
Petechial rush	1	12	95
Petechiae	15	36	98
Purpura	0	0	37
Hematoma	5	0	5
Hematemesis and melena			
Respiratory and circulatory symptoms			
Cough	13	6	31
Dyspnea	10	0	25
Pneumonic infiltrations	13	0	25
Pulmonary edema	8.5	0	5
Shock	21	0	10
Neurological symptoms			
Dizziness	50	12	94
Blurred vision	ND**	12	52
Convulsions	0	0	9
Mortality rate	13	<1	5-20
Total number of cases studied	23	76	125

* From Lahdevirta, 1982 (7).
** ND, no data.
*** From Lee, 1982 (10).

Human serosurvey

Previous serosurveys conducted in 12 counties of Greece in 1981 and 1983 examined 445 and 347 human sera respectively by the IFA test. An additional 1,796 human sera collected from another 9 counties were examined by IFA tests during the years 1984 and 1985. All sera analyzed together and the results summarized according to region and county of origin in which the serosurveys were conducted, as shown in Table 3. The overall antibody prevalence rate was 4%, with a range from 0 to 14%. The maximum percentage of positivity occurred in areas where clinical disease was diagnosed. Moreover, seropositives were detected in 15 of 22 counties, indicating that the virus is widespread in Greece. From the clinical cases and the serosurveys, the ratio of males to females infected is approximately 3:1. The first cases appeared in early May, and cases were observed until late October. The highest risk age group ranged from 35 to 50.

TABLE 3.
Antibody to Hantaan virus in healthy residents of Greece.

Region, county	No. of sera positive/no. tested (% positive)		No. of HFRS cases
Thrace			
Evros	0/129		—
Rodopi	0/62		—
Xanthi	1/79	(1.2)	—
Macedonia			
Serres	0/42		—
Kilkis	6/149	(4.0)	1
Thessaloniki	1/191	(0.5)	—
Halkidiki	6/95	(6.3)	—
Pella	13/93	(14)	5
Imathia	3/185	(1.6)	—
Pieria	2/100	(2.0)	—
Kastoria	0/62		1
Kozani	4/153	(2.6)	1
Thessalia			
Larissa	2/103	(1.9)	—
Karditsa	1/48	(2.0)	1
Magnesia	4/132	(3.0)	—
Trikala	0/0		1
Epirus			
Ioannina	22/282	(7.8)	13
Peloponnisos			
Messinia	0/20		—
Lakonia	1/45	(2.2)	—
Crete			
Hania	0/27		—
Iraklio	3/81	(3.7)	—
Corfu	4/95	(4.2)	—
Others	32/462	(6.9)	—
Total	105/2635	(4)	23

* Blood bank: Residents of examined counties.

Small mammal serosurvey

In 1984 small mammals were collected from an endemic area, Tsepelovo, Epirus, and in 1985 further collections were made in four other areas, one endemic and three nonendemic. *Apodemus flavicollis* was common in fields adjacent to villages and in the surrounding mountains at all collection sites. High antibody titers were detected in four *Apodemus flavicollis* sera (Table 4). Captured house rats (*Rattus rattus*) were common in villages and in a slaughter house of Thessaloniki. Two seropositive *Rattus rattus* were found (Table 4), both from Thessaloniki. To date, neither *Apodemus agrarius*, the host of Hantaan virus, nor *Clethrionomys glareolus*, the host of Puumala virus, was captured at any site sampled. So far, all attempts to isolate virus from lung tissues of *Apodemus flavicollis* and *Rattus rattus* have been unsuccessful.

Serological identification of the virus

Serological identification of the virus responsible for HFRS in Greece was made by examination of patients' sera by both IFA and PRN tests using Hantaan, Seoul, and Puumala viruses. As is shown in Table 5, IFA tests detected high titers to both Hantaan and Seoul viruses, with no significant differences observed between them, and relatively low titers to Puumala virus. In contrast, with PRN tests (Table 6) and the same sera, highest titers were usually observed with Hantaan virus, whereas the reactivity with both Seoul and Puumala viruses was substantially lower than that observed with the IFA test.

DISCUSSION

Hemorrhagic fever with renal syndrome presents a classical and almost unique clinical syndrome whose etiology has now been confirmed by virus isolations and antibody studies. There are, however, prominent differences between the clinical manifestations of the disease in different areas of the world. These differences are believed to be associated with the strain of Hantavirus present. Currently, these clinical forms are described as: Korean hemorrhagic fever, and epidemic hemorrhagic fever, severe forms of the disease associated with prototype Hantaan virus: nephropathia epidemica, a mild form of the disease associated with Puumala virus; and a less severe form of HFRS associated with Seoul virus. In north-

TABLE 4. — Small mammals captured in endemic and nonendemic areas in Greece tested for IFA antibodies to Hantaan virus.

Location	Species	No. of trapped	No. of positives+
Region: Epirus	*Rattus rattus alexandrinus*	41	—
County: Ioannina *	*R. rattus frugivorus*	10	—
Area: Tsepelovo	*Apodemus flavicollis*	23	2
	A. sylvaticus	1	
	Crocidura sp.	1	—
Region: Macedonia			
County: Pella *			
Area: Promahi	*Apodemus flavicollis*	19	2
	A. sylvaticus	4	—
	Mus domesticus	1	—
County: Serres **			
Area: Rice fields	*Apodemus flavicollis*	9	—
County: Thessaloniki			
Area: Rice fields	*Apodemus sylvaticus*	3	—
	A. flavicollis	9	—
County: Thessaloniki **			
Area: Slaughter house	*Rattus rattus alexandrinus*	15	2
Total		136	6

* Endemic « high risk » areas.
** Endemic areas.
+ IFA titers ranging from 1:64 to 1:2048.

TABLE 5.

IFA antibody titers to Hantaan, Seoul, and Puumala viruses in Greek HFRS patients' sera by day of illness.

Serum No.	Day of Illness	Hantaan	Seoul	Puumala
1	9	1024*	1024	16
2	17	8192	8192	256
3	8	8192	8192	256
4	18	4096	4096	256
5	5	1024	1024	<16
6	8	512	512	<16
7	14	512	<6	<16
8	7	1024	256	256
9	8	4096	1024	256
10	9	1024	256	256
11	10	2048	256	64
12	20	2048	512	64
13	10	2048	1024	32
14	28	4096	4096	64
15	10	4096	1024	64
16	9	1024	256	128
17	8	512	128	<16
18	12	8192	2048	128
19	7	4096	2048	128
20	10	4096	256	128
21	Inapparent	256	256	128
Controls				
KHF patient serum		*4096*	256	64
Girard Point Rat serum		256	*512*	16
NE patient serum		128	64	*512*

* Reciprocal of highest dilution showing characteristic cytoplasmic fluorescence.

TABLE 6.

PRNT antibody titers to Hantaan, Seoul and Puumala viruses in Greek patients' sera by day of illness.

Serum No.	Day of Illness	Hantaan	Seoul	Puumala
1	9	2048*	16	32
2	17	1924	16	<16
3	8	64	64	32
4	18	1024	64	<16
5	5	512	64	<16
6	8	128	<16	128
7	14			<32
8	7	4096	64	32
9	8	4096	<16	16
10	9	4096	<16	16
11	10			NT
12	20	128	16	256
13	10	1024	<16	<16
14	28			128
15	10	2048	<16	<16
16	9	NT	NT	NT
17	8	4096	256	1024
18	12	2048	<16	16
19	7	2048	16	16
20	10	4096	16	64
21	Inapparent	4096	64	32
Controls				
KHF patient serum		2048	128	64
Girard Point Rat sera		256	2048	<16
NE patient serum		<16	<16	1024

* Reciprocal of highest dilution yielding ≥50% reduction of plaque dose (100 PFU).

western Europe (Belgium, France, and West Germany) the disease is described as a mild form of HFRS, while in central and southeastern Europe (Hungary, Czechoslovakia, Bulgaria, and Yugoslavia), the disease appears to be severe. A recent report from Bulgaria presents a total of 351 cases of the disease in the past 30 years, with a 14% mortality. This paper identifies a «high risk» area in the mountain of Ropoi, which is a physical part of the official border between northeastern Greece and southern Bulgaria (6).

In Greece, the disease is much more severe than nephropathia epidemica in Scandinavia and HFRS in Western Europe. Among the 23 cases studied, 12 (52%) were severely ill, 8 (35%) required renal dialysis, and 3 (14%) died. Additionally, shock, pulmonary edema, and hemorrhagic manifestations developed in 5, 2, and 3 patients respectively. The most prominent difference between nephropathia epidemica and Korean hemorrhagic fever concerns the mortality rates, which are about 0.5% and 10%, respectively. However, mortality rates are not the only difference. Hemorrhagic manifestation are more frequent in KHF, as are some neurologic symptoms, pulmonary manifestations, and shock. The clinical findings from Greek patients suggest that the disease is more like the severe Asian form of HFRS, caused by Hantaan virus, than the milder nephropathia epidemica caused by Puumala virus.

Sera of the Greek HFRS patients, examined by the IFA test with Hantaan, Puumala, and Seoul viruses, showed equivalent titers against both Hantaan and Seoul viruses, but titers against Puumala virus were much lower, suggesting that this virus was not the infecting agent. In contrast, when the same sera were examined by PRN test with Hantaan, Puumala, and Seoul viruses, they showed little reactivity with Puumala or Seoul viruses, but high titers were seen to Hantaan virus. In general, the 50% reduction point was much higher than the 80% titers, and in many instances 100% reduction was never reached, even in the initial 1:16 dilution. These results suggest that neither Puumala nor Seoul viruses were the infecting agents, and indicate that Hantaan, or a closely related but distinct virus may be the cause of HFRS in Greece.

Serological diagnosis of the disease can be made using the IFA test to detect IgM antibodies during the early days of the disease. The earliest diagnosis we made was on the fifth day of the disease (IgM = 1:2,048, IgG = 1:512), whereas, in a second blood sample taken 15 days later from the same patient, the IgM titer was 1:512 and the IgG titer was 1:16,000.

Hemorrhagic fever with renal syndrome in Greece appears in late May and lasts until early October with a peak in August. The disease is widespread in Greece and may appear as isolated

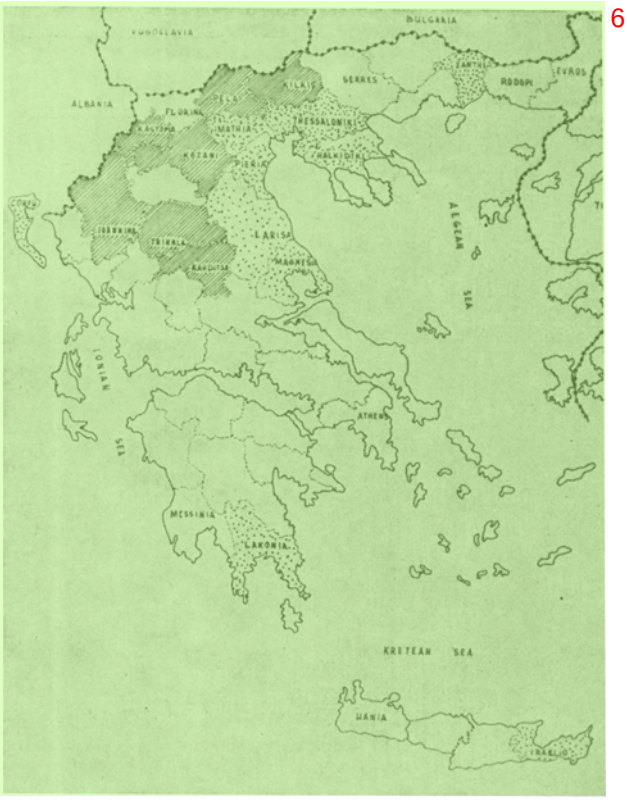


Figure 1 : HFRS distribution in Greece

HFRS cases serologically diagnosed

Antibody to Hantaan virus in healthy residents of Greece

cases or outbreaks, particularly when groups such as woodcutters, and farmers are forced to spend the night outdoors or in temporary quarters, thereby being exposed to infected rodents. Man-to-man transmission apparently does not occur. None of the hospital staff taking care of HFRS patients were found to have antibodies against Hantaan virus. One case only, the wife of a patient, had high antibody titers at the same time as her husband, who was seriously ill. When questioned, however, she denied clinical symptoms in the past year. The infection was inapparent, and the woman may have contacted the infection at the same site as her husband, since she was accompanying her husband to the woods.

Results from the serosurvey screening in the northeastern (Evros), and western (Crete and Corfu) counties revealed that the virus is spread throughout the country. Clinical cases were serologically confirmed in 6 counties. Two endemic areas can be categorized as high risk areas, Tsepelovo and Promahi. Additionally, 15 of 22 counties, where seropositives were found, can also be characterized as endemic areas of the virus (Fig. 1). The percentage of seropositives in the total

number of sera examined was 4%, while the percentage in endemic areas was 14%. The age groups of 30 to 60 years were most frequently positive, with a significantly higher incidence among males. The male to female ratio was approximately 3:1.

The vertebrate host of the virus causing severe HFRS in Greece may be *Apodemus flavicollis*. Antibodies to Hantaan virus have been found only in this species, and *Apodemus flavicollis* is found throughout the Balkan Peninsula.

In conclusion, HFRS is endemic in Greece, and the clinical manifestations of the disease more closely resemble Korean hemorrhagic fever than nephropathia epidemica or the western European form of HFRS. There is a broad spectrum of symptoms, and the infection ranges from inapparent to lethal. Further attempts must be made to isolate the virus from patients and rodents. The epidemiology of HFRS in the Balkan Peninsula has not yet been fully elucidated.

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